

05

Inverse of a Matrix

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A square matrix's determinant is nonzero — the inverse transformation of a geometric operation

5.1 Inverse of a Matrix



Core skills you'll master in this section:

- ▶ Invertible matrix: a square matrix whose determinant is nonzero.
- ▶ A matrix is a linear transformation; its inverse matrix is that transformation's "inverse operation," attempting to restore the original geometric structure. The determinant reflects the change in area.
- ▶ A square matrix is invertible: its geometric transformation does not cause dimensional collapse or loss of information.
- ▶ The inverse of a scaling matrix restores the original position via inversely proportional scaling.
- ▶ The inverse of a rotation is a rotation by the opposite angle, and rotation matrices are orthogonal matrices.
- ▶ The inverse of a shear transformation restores the shape by canceling out the tilt component.
- ▶ The inverse of a mirror transformation is itself.
- ▶ Orthogonal matrices include rotations, mirrors, permutations, the identity matrix, and more.

How to compute a matrix inverse by hand is, in the author's view, the most boring part of the matrix inverse. The computational formula itself does little to explain the essence of the matrix inverse, or its role in mathematics and the real world.

To understand the matrix inverse more intuitively, we still need to bring out that powerful tool — the geometric perspective!

As mentioned earlier in this book, matrix multiplication is essentially a linear transformation, capable of performing rotation, scaling, shearing, projection, and other operations.

The inverse of a matrix, then, attempts to run these transformations in reverse and restore the original geometric structure.

For example, the inverse of a rotation matrix is a rotation in the opposite direction, and the inverse of a scaling matrix is scaling by the reciprocal ratio. If a transformation causes information loss, however, the transformation is not invertible.

In other words, whether a matrix is invertible depends on whether it preserves the full information of the space. In addition, the matrix inverse is closely related to concepts such as the determinant, rank, and linear dependence.

In this section, we'll start from a geometric perspective to explore the meaning of the matrix inverse, and analyze how different types of linear transformations affect a matrix's invertibility.

What Is an Inverse Matrix

First, the precondition for matrix A to have an inverse is that A must be a square matrix, that is, its number of rows equals its number of columns.

⚠️ Note that not every square matrix has an inverse matrix.

For an $n \times n$ square matrix A , if there exists a matrix A^{-1} such that:

$$A @ A^{-1} = A^{-1} @ A = I$$

where I is the $n \times n$ identity matrix, then we call the square matrix A an invertible matrix, or a non-singular matrix.

Geometric Perspective

As mentioned earlier in this book, for an $n \times n$ square matrix A , $Ax = y$ can be viewed as some geometric transformation, such as scaling, rotation, shearing, mirroring, or a sequential combination of these.

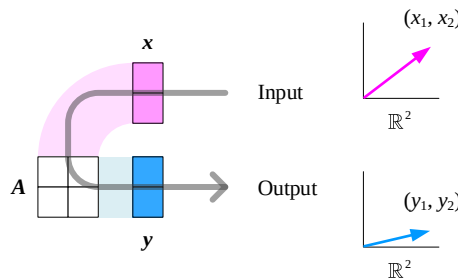


Figure 1. The Column Vector x Under the 2×2 Matrix A Maps to the Column Vector y

As shown in Figure 2, $A^{-1}y = x$ can be regarded as the inverse of these geometric transformations. Of course, this holds only if matrix A is invertible; otherwise, matrix A 's geometric transformation cannot be run in reverse.

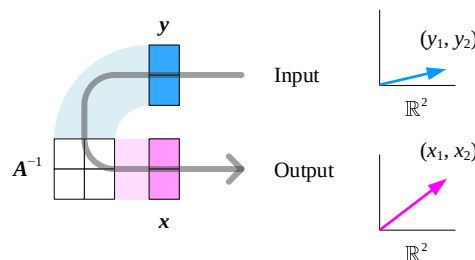


Figure 2. The Column Vector y Under the 2×2 Matrix A^{-1} Maps to the Column Vector x

Existence Condition for the Matrix Inverse

Take the 2×2 matrix as an example

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

its inverse matrix is:

$$\mathbf{A}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

As long as $ad - bc \neq 0$, the inverse can be found.

As you've probably noticed, $ad - bc$ is exactly $\mathbf{A}$'s determinant $\det(\mathbf{A})$!

As mentioned earlier in this book, the determinant represents area or volume; if matrix $\mathbf{A}$'s determinant is 0, this means that after the linear transformation by matrix $\mathbf{A}$, the geometric shape "collapses"! This causes some information to be lost, so the matrix is not invertible.

If $\det(\mathbf{A}) \neq 0$, then after the transformation the geometric shape will not collapse, no information is lost, and therefore the transformation is invertible.

Code 1 A custom function computes the inverse of a 2×2 matrix. Let's go over the key statements below.

a Defines a function named `inverse_2x2_A`. The function computes the inverse of a 2×2 matrix, and checks whether the matrix is invertible. If the matrix's determinant is 0, the function prints a message and returns `None`, indicating that the matrix is not invertible. `def` is the keyword used to define a new function; `inverse_2x2_A` is the function's name; `A` is the function's input parameter, representing a 2×2 matrix. The input `A` needs to be a 2×2 nested list or a numpy array.

b First, extract the two elements `a` and `b` from the first row of matrix `A`; `A[0]` represents the first row of matrix `A`, and `a, b = A[0]` assigns the two values of this row to `a` and `b`, respectively. Then extract the two elements `c` and `d` from the second row of `A`; this works similarly to the line above, except that here `A[1]` represents the second row of matrix `A`.

c Computes the determinant of `A`; the determinant is a value used to determine whether a matrix is invertible. If it equals 0, the matrix is not invertible.

d This line is an if conditional: if the value of `det` equals 0, the code block below is executed. The line `return None` is part of the if block; if `det == 0`, the function immediately returns `None`, meaning the function ends here without further computation.

e This line creates a 2×2 numpy array with elements `d, -b, -c, a`, then divides the entire matrix by `det` to obtain the inverse of `A`, storing the result in the variable `inv_A`.

f Creates a matrix and calls the custom function to compute its inverse.

Code 1. Inverse of a 2×2 Matrix |  LA_Ch05_01_01.ipynb

```

## 初始化
import numpy as np

## 自定义函数

def inverse_2x2_A(A):
    """
    A = [[a, b],
          [c, d]]
    """
    # 提取矩阵元素
    a, b = A[0]
    c, d = A[1]

    # 计算行列式 det(A) = ad - bc
    det = a * d - b * c

    # 判断行列式是否为零
    if det == 0:
        print("矩阵不可逆，行列式为零")
        return None

    # 计算逆矩阵
    inv_A = np.array([[d, -b], [-c, a]]) / det
    return inv_A

## 定义矩阵
A = [[1, 2],
      [3, 4]]
A = np.array(A)

## 计算逆矩阵
inverse_2x2_A(A)

```



LA_05_01_02.ipynb shows how to compute a matrix inverse using `numpy.linalg.inv()`, and LA_05_01_03.ipynb introduces computing a matrix inverse using SymPy symbolic computation; please study these on your own.

Next, let's look at a few concrete examples of matrix inverses.

Scaling and Inverse Scaling

The inverse of a scaling matrix is scaling by the reciprocal ratio.

Given the specific scaling matrix S as

$$\mathbf{S} = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}$$

The effect of the square matrix $\mathbf{S}$ is to scale vectors in two-dimensional space along the x_1 axis and the x_2 axis, respectively.

Specifically, the square matrix $\mathbf{S}$ will enlarge the coordinate in the x_1 direction by a factor of 1.5, that is, $3/2$; meanwhile, it will shrink the coordinate in the x_2 direction to $2/3$ of its original value.

This means that all points are stretched horizontally and compressed vertically, causing the entire shape to deform.

However, if we compute the determinant of the square matrix $\mathbf{S}$, we find that $\det(\mathbf{S}) = 1$; this means that this scaling matrix does not cause any change in area.

The Inverse of the Square Matrix $\mathbf{S}$

$$\mathbf{S}^{-1} = \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}$$

The inverse matrix of the square matrix $\mathbf{S}$ has exactly the opposite effect.

Each scaling factor of the inverse matrix $\mathbf{S}^{-1}$ is the reciprocal of the corresponding scaling factor of the original matrix; it therefore shrinks the coordinate in the x_1 direction to $2/3$, while enlarging the coordinate in the x_2 direction by a factor of 1.5, completely canceling out the original matrix's transformation, so that any point transformed by the original matrix can be restored to its original position.

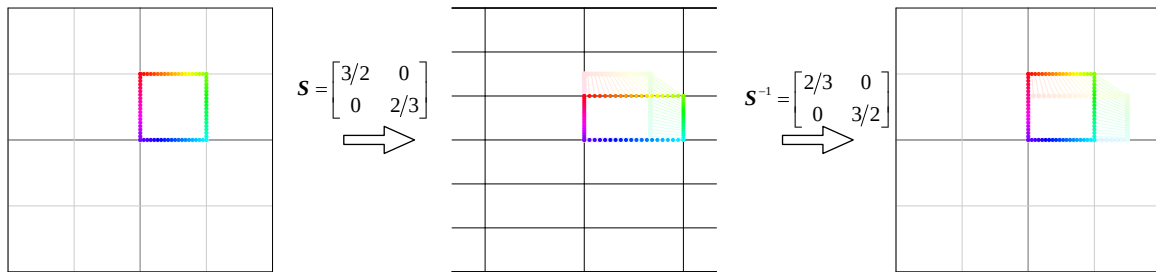


Figure 3. Scaling and the Inverse Scaling Operation in the Plane, 2×2 Matrix

Compute the Product of $\mathbf{S}$ and $\mathbf{S}^{-1}$

$$\mathbf{S} @ \mathbf{S}^{-1} = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix} @ \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiplying the two together gives the identity matrix; the identity matrix means no change.

Conversely, compute the product of $\mathbf{S}^{-1}$ and $\mathbf{S}$

$$\mathbf{S}^{-1} @ \mathbf{S} = \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix} @ \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The inverse of a diagonal square matrix has the same structure, except that each nonzero entry on the main diagonal is the reciprocal of the corresponding entry in the original matrix.

Therefore, as long as the entries on the diagonal are nonzero, any diagonal square matrix is invertible.

Rotation and Inverse Rotation

The inverse of a rotation matrix is a rotation in the opposite direction.

Given the rotation matrix $\mathbf{R}$

$$\mathbf{R} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$$

The effect of the square matrix $\mathbf{R}$ is to rotate all vectors in two-dimensional space counterclockwise by 120° about the origin. That is, the origin (the zero vector) stays fixed – after rotation, the zero vector $\mathbf{0}$ is still $\mathbf{0}$.

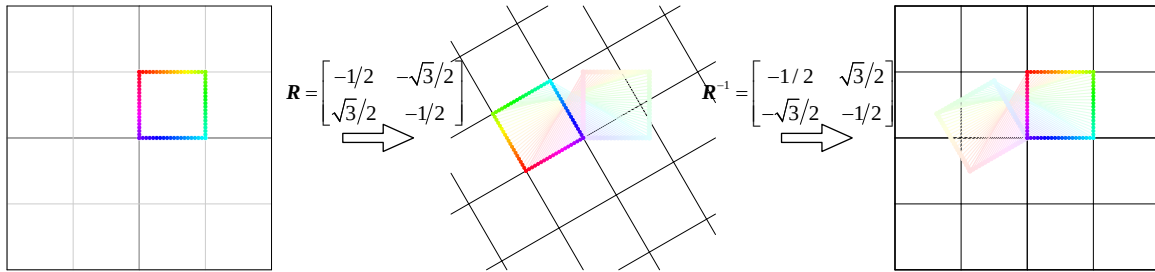


Figure 4. Rotation and the Inverse Rotation Operation in the Plane, 2×2 Matrix

Compute the Determinant of the Square Matrix $\mathbf{R}$

$$\det(\mathbf{R}) = (-1/2) \times (-1/2) - (-\sqrt{3}/2) \times (\sqrt{3}/2) = 1$$

The determinant's value is 1, which means this transformation does not change area.

The square matrix $\mathbf{R}$'s inverse matrix $\mathbf{R}^{-1}$ has the same structure, but with the opposite angle

$$\mathbf{R}^{-1} = \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}$$

Since the square matrix $\mathbf{R}$ represents a counterclockwise rotation of 120° , its inverse matrix $\mathbf{R}^{-1}$ should represent a clockwise rotation of 120° .

Compute the Product of $\mathbf{R}$ and $\mathbf{R}^{-1}$

$$\mathbf{R} @ \mathbf{R}^{-1} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix} @ \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

If a point is rotated counterclockwise by 120° by the square matrix $\mathbf{R}$, and then rotated clockwise by 120° by multiplying by $\mathbf{R}^{-1}$, it returns to its original position.

Please compute the matrix product of $\mathbf{R}^{-1}$ and $\mathbf{R}$

In addition, for rotation matrices, the following relationship is worth noting

$$\mathbf{R}^T = \mathbf{R}^{-1}$$

That is,

$$\mathbf{R}^T \mathbf{R} = \mathbf{I}, \quad \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

A square matrix that satisfies the above relationship is called an orthogonal matrix.



Later in this book, we will discuss orthogonal matrices in detail.

Shear and Inverse Shear

The effect of a shear matrix is to tilt the coordinate system along a certain direction, so that one component of a vector is stretched or compressed by the other component, without changing the shape's area. The inverse of a shear matrix restores the original shape through an inverse shear.

Given the shear matrix:

$$\mathbf{K} = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

The effect of the square matrix $\mathbf{K}$ is to apply a shear transformation along the x_1 direction to all vectors in two-dimensional space.

Specifically: a vector's component in the x_2 direction remains unchanged.

A vector's component in the x_1 direction increases by 0.5 times its x_2 coordinate. The point (1, 1) becomes (1.5, 1) after the transformation. The point (2, 3) becomes (3.5, 3) after the transformation. Note that the origin (0, 0) stays fixed, since it remains itself after shearing.

Compute the Determinant of the Square Matrix $\mathbf{K}$

$$\det(\mathbf{K}) = (1 \times 1) - (0.5 \times 0) = 1$$

The determinant's value is 1, which means this transformation does not change area. It only changes the shape, without stretching or compressing the overall size of the space.

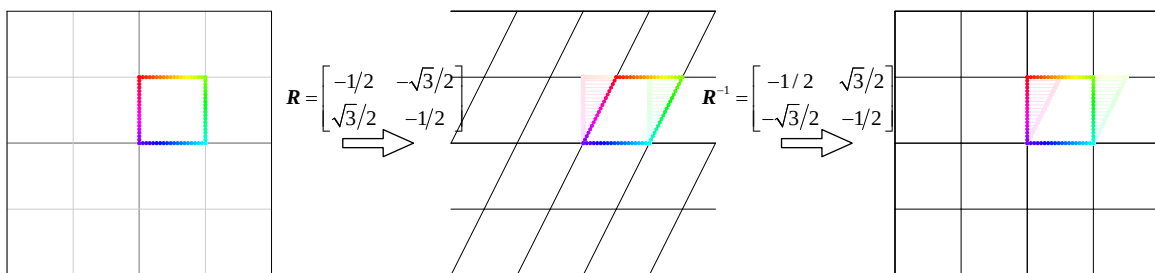


Figure 5. Shearing and the Inverse Shear Operation in the Plane, 2×2 Matrix

The shear transformation is invertible; the square matrix $\mathbf{K}$'s inverse matrix $\mathbf{K}^{-1}$ also has a similar structure.

Since the shear operation simply adds 0.5 times the x_1 coordinate to the x_2 direction, the inverse transformation should subtract 0.5 times the x_2 coordinate, thereby canceling out the shear effect. Therefore, the inverse of the shear matrix is:

$$\mathbf{K}^{-1} = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix}$$

The effect of this inverse matrix is: the x_1 coordinate decreases by 0.5 times the x_2 coordinate, canceling out the previous shear transformation; the x_2 coordinate remains unchanged.

Compute the Product of $\mathbf{K}$ and $\mathbf{K}^{-1}$

$$\mathbf{K} @ \mathbf{K}^{-1} = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

It's worth noting that $\mathbf{K}$ and $\mathbf{K}^{-1}$ are both upper triangular matrices. The inverse of an upper triangular matrix is still an upper triangular matrix.

Mirroring and Inverse Mirroring

The effect of a mirror matrix is to reflect symmetrically about a certain axis.

Given the mirror matrix:

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

The effect of the square matrix $\mathbf{M}$ is to reflect all vectors in two-dimensional space along the x_2 axis (the vertical axis).

Specifically: the x_1 coordinate changes sign — the square matrix $\mathbf{M}$ negates every point's coordinate in the x_1 direction, that is, it flips horizontally.

The x_2 coordinate remains unchanged — the square matrix $\mathbf{M}$ does not change the coordinate in the x_2 direction.

For example, the point (1, 1) becomes (-1, 1) after the transformation; the point (-1, 0) becomes (1, 0) after the transformation.

The origin (0, 0) stays fixed, since it remains itself after reflection. That is, the zero vector $\mathbf{0}$ is still the zero vector $\mathbf{0}$ after mirroring.

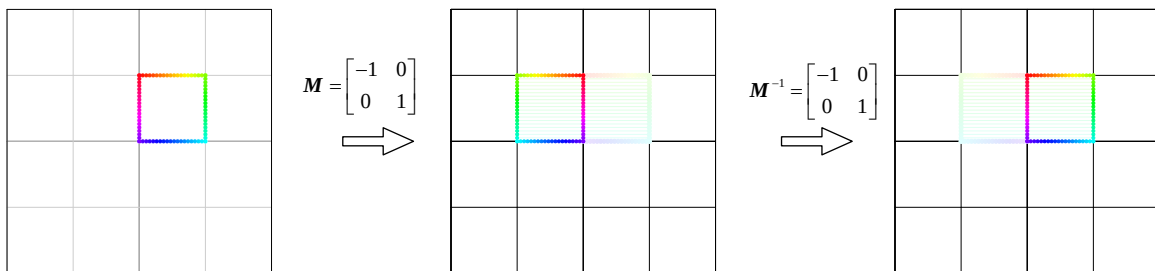


Figure 6. Mirroring and the Inverse Mirroring Operation in the Plane, 2×2 Matrix

Compute the Determinant of the Square Matrix $\mathbf{M}$

$$\det(\mathbf{M}) = (-1) \cdot (1) - (0) \cdot (0) = -1$$

The absolute value of the determinant is 1, which means this transformation does not change area; but since the determinant is negative, it means the transformation changes orientation, that is, it flips the orientation of the space.

For example, viewed counterclockwise, before the transformation $\mathbf{e}_1$ leads $\mathbf{e}_2$; after the $\mathbf{M}$ transformation, $\mathbf{e}_2$ leads $\mathbf{e}_1$.

The mirror transformation is invertible; the square matrix $\mathbf{M}$'s inverse matrix $\mathbf{M}^{-1}$ is also a mirror matrix

$$\mathbf{M}^{-1} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Because a mirror operation is symmetric, performing the same mirror operation a second time restores the original position.

Compute the Product of $\mathbf{M}$ and $\mathbf{M}^{-1}$

$$\mathbf{M} @ \mathbf{M}^{-1} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

That is to say, the inverse of a mirror matrix is itself!

That is,

$$\mathbf{M} = \mathbf{M}^{-1}$$

This matches intuition — if a point is flipped once by a mirror transformation, performing the same mirror transformation again returns it to its original position.

In addition, a mirror matrix is an orthogonal matrix, because,

$$\mathbf{M}^T \mathbf{M} = \mathbf{I}, \quad \mathbf{M} \mathbf{M}^T = \mathbf{I}$$

that is, its transpose equals its inverse.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please determine whether the following matrices are invertible.

▶ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Q2. Please compute the inverse of the following scaling matrices, and describe the matrix's geometric operation.

▶ $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

▶ $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

Q3. Please compute the inverse of the following shear matrices, and describe the matrix's geometric operation.

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

Q4. Please compute the inverse of the following rotation matrices (about the origin), and describe the matrix's geometric operation.

▶ $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$

▶ $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Q5. Please compute the inverse of the following mirror matrices, and describe the matrix's geometric operation.

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$


$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$


$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$